

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **4-Bromo-2,5-difluorobenzenesulfonyl chloride**  
 Product Description: **4-Bromo-2,5-difluorobenzenesulfonyl chloride**  
 Cat No. : 399880000, 399880010, 399880050  
 CAS No 207974-14-9  
 Molecular Formula C<sub>6</sub> H<sub>2</sub> Br Cl F<sub>2</sub> O<sub>2</sub> S

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                           |                     |
|---------------------------|---------------------|
| Skin Corrosion/Irritation |                     |
|                           | Category 1 B (H314) |

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H314 - Causes severe skin burns and eye damage

**Precautionary Statements**

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## Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

EUH029 - Contact with water liberates toxic gas

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                             | CAS No      | Weight % |
|-------------------------------------------------------|-------------|----------|
| 4-Bromo-2,5-difluorobenzene-1-sulfonyl chloride, 95%+ | 207974-14-9 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. . Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Contact with water liberates toxic gas.

### **Hazardous Combustion Products**

Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen fluoride, Hydrogen chloride gas.

### Advice for fire-fighters

Thermal decomposition can lead to release of irritating gases and vapors. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not allow contact with water. Avoid dust formation. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

### Specific End Uses

Use in laboratories.

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## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### **Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

#### **Physical State**

Solid

#### **Odor**

No information available

#### **Odor Threshold**

No data available

#### **pH**

No information available

#### **Melting Point/Range**

37 - 39 °C / 98.6 - 102.2 °F

#### **Softening Point**

No data available

#### **Boiling Point/Range**

No information available

#### **Flash Point**

No information available

**Method -** No information available

#### **Evaporation Rate**

Not applicable

Solid

#### **Flammability (solid,gas)**

No information available

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**Explosion Limits** No data available

|                                     |                          |       |
|-------------------------------------|--------------------------|-------|
| <b>Vapor Pressure</b>               | No data available        |       |
| <b>Vapor Density</b>                | Not applicable           | Solid |
| <b>Specific Gravity / Density</b>   | No data available        |       |
| <b>Bulk Density</b>                 | No data available        |       |
| <b>Water Solubility</b>             | No information available |       |
| <b>Solubility in other solvents</b> | No information available |       |

**Partition Coefficient (n-octanol/water)**

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Autoignition Temperature</b>  | No data available        |       |
| <b>Decomposition Temperature</b> | No data available        |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

|                          |                     |
|--------------------------|---------------------|
| <b>Molecular Formula</b> | C6 H2 Br Cl F2 O2 S |
| <b>Molecular Weight</b>  | 291.5               |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Water reactive. Moisture sensitive.

### Possibility of Hazardous Reactions

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>      | Contact with water liberates toxic gas.  |

### Conditions to Avoid

Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture. Avoid dust formation.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen fluoride. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

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## Information on Toxicological Effects

**Product Information** No acute toxicity information is available for this product

**(a) acute toxicity;**

Oral

No data available

Dermal

No data available

Inhalation

No data available

**(b) skin corrosion/irritation;**

Category 1 B

**(c) serious eye damage/irritation;**

No data available

**(d) respiratory or skin sensitization;**

Respiratory

No data available

Skin

No data available

**(e) germ cell mutagenicity;**

No data available

**(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

No data available

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

Target Organs

No information available.

**(j) aspiration hazard;**

Not applicable

Solid

**Other Adverse Effects**

The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed**

Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

Do not empty into drains.

**Persistence and degradability**

No information available

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|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <u>Bioaccumulative potential</u>       | No information available                                                  |
| <u>Mobility in soil</u>                | No information available.                                                 |
| <u>Endocrine Disruptor Information</u> | This product does not contain any known or suspected endocrine disruptors |
| <u>Other adverse effects</u>           | No information available                                                  |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                           |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u>             |                                                                                                                                                                                                           |
| <u>Waste from Residues/Unused Products</u> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations                                       |
| <u>Contaminated Packaging</u>              | Empty containers should be taken to an approved waste handling site for recycling or disposal Dispose of this container to hazardous or special waste collection point.                                   |
| <u>Other Information</u>                   | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                                                                          |
|-------------------------------------|------------------------------------------------------------------------------------------|
| <u>IMDG/IMO</u>                     |                                                                                          |
| UN-No                               | UN3261                                                                                   |
| Hazard Class                        | 8                                                                                        |
| Packing Group                       | II                                                                                       |
| Proper Shipping Name                | Corrosive solid, acidic, organic, n.o.s. 4-Bromo-2,5-difluorobenzene-1-sulfonyl chloride |
| <u>Road and Rail Transport</u>      |                                                                                          |
| UN-No                               | UN3261                                                                                   |
| Hazard Class                        | 8                                                                                        |
| Packing Group                       | II                                                                                       |
| Proper Shipping Name                | Corrosive solid, acidic, organic, n.o.s. 4-Bromo-2,5-difluorobenzene-1-sulfonyl chloride |
| <u>IATA</u>                         |                                                                                          |
| UN-No                               | UN3261                                                                                   |
| Hazard Class                        | 8                                                                                        |
| Packing Group                       | II                                                                                       |
| Proper Shipping Name                | Corrosive solid, acidic, organic, n.o.s. 4-Bromo-2,5-difluorobenzene-1-sulfonyl chloride |
| <u>Special Precautions for User</u> | No special precautions required                                                          |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                           |            |
|---------------------------|------------|
| International Inventories | X = listed |
|---------------------------|------------|

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## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

22-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**